

The division identity

For a nonzero polynomial divisor, the dividend equals divisor times quotient plus remainder.

$$P(x) = D(x)Q(x) + R(x)$$

How could this identity check a division result?

The remainder's degree

The remainder is zero or has degree smaller than the divisor.

What forms can a remainder have when the divisor has degree 2?

Dividing by a monomial

$$\frac{6x^4 - 9x^3 + 3x^2}{3x^2}$$

What polynomial is equivalent for nonzero x ?

An exact division

$$\frac{x^2 + 7x + 10}{x + 2}$$

What quotient and remainder result from long division?

A nonzero remainder

$$\frac{x^2 + 2x + 5}{x + 1}$$

What quotient and remainder result from long division?

Writing a quotient expression

$$P(x) = (x - 2)(x + 5) + 3$$

How can $P(x)/(x - 2)$ be written for x not equal to 2?

Missing powers

$$\frac{x^3 - 8}{x - 2}$$

Which zero coefficients belong in the division setup?

Dividing a cubic

$$\frac{2x^3 - 3x^2 - 11x + 6}{x - 3}$$

What quotient and remainder result?

A non-monic divisor

$$\frac{4x^2 + 7x + 2}{2x + 1}$$

What quotient and remainder result from long division?

A quadratic divisor

$$\frac{x^4 + 2x^2 + 3}{x^2 + 1}$$

What quotient and remainder result from long division?

Checking a division

A proposed result has quotient $x + 3$ and remainder 2.

$$\frac{x^2 + 2x - 1}{x - 1}$$

Does multiplication verify the proposed result?

Synthetic division

The standard synthetic-division setup divides by $x - k$.

What value of k belongs in the setup for division by $x + 4$?

Synthetic division with a missing term

$$\frac{x^3 + 2x - 3}{x - 1}$$

What coefficient list should be used?

Performing synthetic division

$$\begin{array}{r} x^3 - 2x^2 - 5x + 6 \\ \hline x - 1 \end{array}$$

What quotient and remainder result?

Interpreting the final row

Synthetic division by $x - 2$ gives quotient coefficients 1, 3 and remainder -4.

What identity connects the original polynomial, quotient, and remainder?

Choosing a division method

$$\frac{x^4 - 1}{x^2 + 1}$$

Does the standard x - k synthetic setup apply to this divisor?

A lower-degree dividend

$$\frac{2x + 3}{x^2 + 1}$$

What are the polynomial quotient and remainder?

A factor test

Division by $x + 2$ gives remainder 0.

What does this tell us about $x + 2$ as a factor?

The division-domain restriction

$$\frac{x^3 - 8}{x - 2}$$

If the quotient is a polynomial, is the original rational expression defined at $x = 2$?